

## LYDENBURG, South Africa

WMO: 681850

Lat: 25.1119S

Lon: 30.4767E

Elev: 1433

StdP: 85.25

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 2.6        | 3.9 | -8.9                          | 2.1 | 13.7 | -6.8 | 2.5 | 13.0 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | 0.325 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 8.8                    | 29.3            | 16.7 | 28.1 | 16.7 | 27.1 | 16.9 | 20.3                | 25.2 | 19.7 | 24.5 | 19.2 | 23.9 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.0                            | 16.4 | 22.5 | 18.2 | 15.7 | 21.9 | 17.6 | 15.1 | 21.5 | 65.7          | 25.2 | 63.5 | 24.6 | 61.3 | 23.9 | 21.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| N/A               | N/A  | N/A | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.4   | 21.0  | 20.8  | 19.6  | 17.4  | 15.0  | 12.8  | 12.1  | 14.8  | 17.5  | 18.4  | 19.1  | 20.2 |
|                                                                                  | DBStd       | 3.76   | 1.62  | 1.51  | 2.02  | 2.20  | 2.34  | 2.34  | 2.37  | 2.71  | 3.28  | 2.78  | 2.75  | 2.01 |
|                                                                                  | HDD10.0     | 16     | 0     | 0     | 0     | 0     | 2     | 4     | 8     | 1     | 1     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 743    | 1     | 1     | 11    | 41    | 105   | 166   | 193   | 111   | 52    | 34    | 23    | 6    |
|                                                                                  | CDD10.0     | 2705   | 340   | 303   | 297   | 223   | 156   | 89    | 73    | 151   | 226   | 259   | 273   | 316  |
|                                                                                  | CDD18.3     | 391    | 83    | 71    | 49    | 14    | 1     | 0     | 0     | 2     | 27    | 35    | 46    | 63   |
|                                                                                  | CDH23.3     | 2029   | 334   | 284   | 170   | 66    | 24    | 6     | 4     | 73    | 296   | 264   | 221   | 285  |
|                                                                                  | CDH26.7     | 310    | 48    | 41    | 17    | 4     | 1     | 0     | 4     | 63    | 52    | 33    | 48    |      |
| Wind                                                                             | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Precipitation                                                                    | PrecAvg     | 829    | 141   | 125   | 103   | 46    | 15    | 7     | 7     | 9     | 22    | 79    | 128   | 148  |
|                                                                                  | PrecMax     | 1286   | 271   | 350   | 186   | 91    | 41    | 29    | 26    | 27    | 81    | 138   | 253   | 262  |
|                                                                                  | PrecMin     | 437    | 44    | 38    | 24    | 4     | 0     | 0     | 1     | 1     | 2     | 38    | 48    | 62   |
|                                                                                  | PrecStd     | 191    | 62    | 84    | 44    | 21    | 13    | 8     | 7     | 8     | 19    | 27    | 44    | 46   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 30.2  | 30.0  | 28.7  | 27.1  | 25.3  | 24.3  | 23.7  | 27.2  | 30.4  | 30.2  | 29.4  | 30.1 |
|                                                                                  |             | MCWB   | 18.2  | 18.1  | 17.5  | 15.5  | 13.4  | 11.7  | 10.2  | 13.6  | 13.9  | 15.4  | 16.8  | 18.2 |
|                                                                                  | 2%          | DB     | 27.9  | 28.1  | 26.8  | 25.1  | 23.8  | 22.0  | 21.5  | 25.5  | 28.4  | 28.2  | 27.5  | 28.0 |
|                                                                                  |             | MCWB   | 19.3  | 18.5  | 17.6  | 15.7  | 13.0  | 10.7  | 9.6   | 12.1  | 13.5  | 15.4  | 16.9  | 18.0 |
|                                                                                  | 5%          | DB     | 26.6  | 26.6  | 25.4  | 23.7  | 22.3  | 20.2  | 19.5  | 23.8  | 26.7  | 26.3  | 25.9  | 26.4 |
|                                                                                  |             | MCWB   | 18.7  | 18.1  | 17.5  | 15.4  | 12.5  | 10.2  | 9.1   | 11.7  | 13.4  | 15.3  | 16.8  | 18.4 |
|                                                                                  | 10%         | DB     | 25.2  | 25.2  | 23.9  | 22.1  | 20.5  | 18.3  | 17.5  | 21.5  | 24.7  | 24.4  | 24.2  | 24.9 |
|                                                                                  |             | MCWB   | 18.6  | 17.8  | 17.2  | 15.0  | 11.9  | 9.7   | 8.7   | 10.9  | 12.9  | 15.2  | 16.9  | 18.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 21.1  | 20.8  | 20.9  | 18.2  | 15.3  | 12.7  | 11.8  | 15.1  | 16.6  | 19.4  | 20.5  | 20.7 |
|                                                                                  |             | MCDB   | 27.0  | 26.6  | 23.8  | 23.6  | 22.7  | 20.3  | 18.5  | 23.3  | 26.5  | 23.2  | 23.4  | 26.3 |
|                                                                                  | 2%          | WB     | 20.2  | 19.7  | 19.6  | 17.3  | 14.2  | 11.8  | 10.7  | 13.7  | 15.5  | 18.1  | 19.3  | 19.9 |
|                                                                                  |             | MCDB   | 25.8  | 25.3  | 23.2  | 22.0  | 21.0  | 19.6  | 18.4  | 21.2  | 24.8  | 23.1  | 22.9  | 24.9 |
|                                                                                  | 5%          | WB     | 19.7  | 19.1  | 18.8  | 16.6  | 13.4  | 11.1  | 9.9   | 13.0  | 14.7  | 17.2  | 18.5  | 19.1 |
|                                                                                  |             | MCDB   | 24.9  | 24.5  | 22.8  | 21.1  | 19.6  | 18.0  | 17.3  | 20.2  | 23.3  | 22.2  | 22.8  | 23.8 |
|                                                                                  | 10%         | WB     | 19.1  | 18.5  | 18.1  | 15.9  | 12.6  | 10.3  | 9.2   | 12.1  | 13.7  | 16.3  | 17.8  | 18.6 |
|                                                                                  |             | MCDB   | 24.0  | 23.5  | 22.1  | 20.3  | 18.4  | 16.6  | 15.7  | 19.1  | 21.9  | 21.6  | 22.3  | 23.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.8   | 9.6   | 9.2   | 10.5  | 13.0  | 13.3  | 13.4  | 13.9  | 14.0  | 11.4  | 9.2   | 8.9  |
|                                                                                  |             | MCDBR  | 11.5  | 12.1  | 12.0  | 12.6  | 14.6  | 15.2  | 15.3  | 16.3  | 16.5  | 14.7  | 12.5  | 12.1 |
|                                                                                  |             | MCWBR  | 4.6   | 4.2   | 4.4   | 4.6   | 5.8   | 6.3   | 6.4   | 6.3   | 6.0   | 5.1   | 4.4   | 4.5  |
|                                                                                  | 5% WB       | MCDBR  | 9.5   | 10.4  | 9.5   | 10.3  | 12.7  | 13.3  | 13.7  | 13.9  | 14.5  | 11.3  | 9.5   | 9.5  |
| MCWBR                                                                            |             | 4.3    | 4.4   | 4.2   | 4.6   | 5.6   | 5.9   | 6.2   | 5.8   | 5.9   | 4.8   | 4.3   | 4.1   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.375  | 0.369 | 0.363 | 0.339 | 0.305 | 0.288 | 0.289 | 0.344 | 0.393 | 0.418 | 0.384 | 0.387 |      |
|                                                                                  | taud        | 2.401  | 2.417 | 2.421 | 2.451 | 2.508 | 2.534 | 2.516 | 2.351 | 2.244 | 2.220 | 2.353 | 2.361 |      |
|                                                                                  | Ebn at Noon | 968    | 959   | 936   | 914   | 907   | 905   | 916   | 893   | 890   | 901   | 953   | 959   |      |
|                                                                                  | Edh at Noon | 128    | 123   | 116   | 103   | 89    | 82    | 87    | 112   | 137   | 148   | 133   | 133   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.18   | 6.04  | 5.38  | 4.62  | 4.32  | 3.92  | 4.19  | 4.86  | 5.54  | 5.67  | 5.94  | 6.11  |      |
|                                                                                  | RadStd      | 0.49   | 0.56  | 0.47  | 0.36  | 0.21  | 0.23  | 0.13  | 0.26  | 0.33  | 0.39  | 0.53  | 0.41  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page